

Jumpstarting Q-Learning with Suboptimal Pathfinding: A Hybrid Approach for Faster Convergence in Pacman

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Introduction

Reinforcement Learning (RL) teaches agents to make optimal decisions through trial and error. However, traditional Q-learning often suffers from slow convergence, especially in complex environments. In this study, we explore a hybrid method that initializes Q-values using demonstration trajectories from a suboptimal search-based agent. By leveraging prior (though imperfect) knowledge, we aim to accelerate learning and enhance overall performance. This approach has potential applications in robotics and game AI especially when learning from scratch is impractical or inefficient.

Research Goal

To evaluate whether initializing Q-learning with Q-values derived from a suboptimal pathfinding agent leads to **faster convergence**, **higher average score** and **higher win rate** compared to standard (cold-start) Q-learning.

Methods

We implemented Q-learning in the UC Berkeley Pacman AI environment using three agent configurations that differ only in their Q-value initialization strategy:

Cold-start Q-Learning: Standard Q-learning with zero-initialized Q-values

Warm-start Q-Learning (DFS): Q-values initialized using trajectories from the DepthFirstSearchAgent

Warm-start Q-Learning (ClosestDotSearch): Q-values initialized using trajectories from the Hyperparameters:

$\gamma = 1.0$ (discount factor)

$\alpha = 0.2$ (learning rate)

$\epsilon = 0.05$ (exploration rate)

Convergence was measured by:

Max Q-value change: $\delta < 0.01$

Matrix residual norm: $||Q_{\text{new}} - Q_{\text{old}}|| < 0.1$

All agents were trained over **1000 episodes** on layouts including *smallGrid* and *mediumClassic*.

Figures and Results

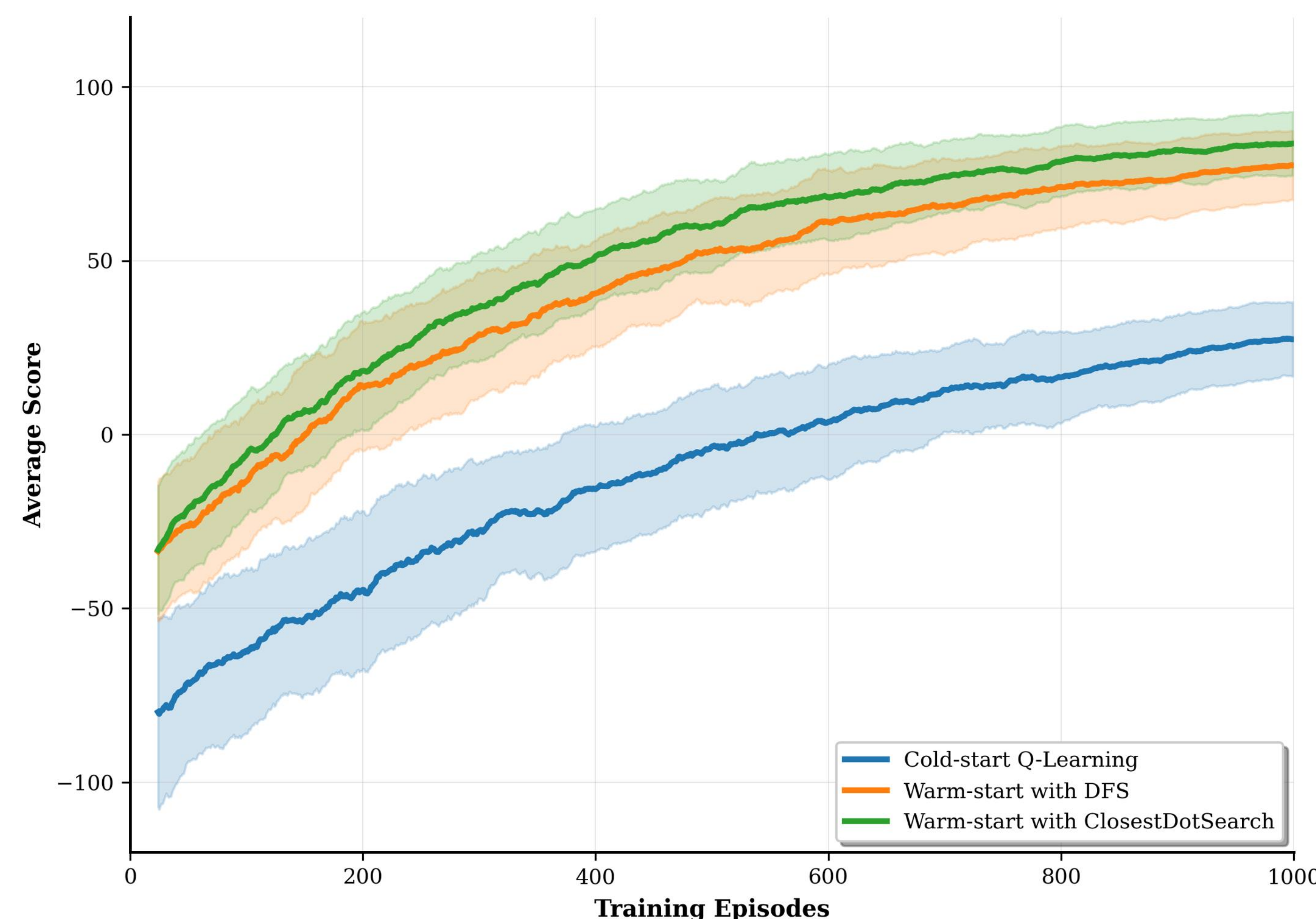


Fig. 1: Score vs. Training Episodes

The warm-start Q-learning agents (orange, green) consistently outperform cold-start (blue), achieving higher scores and faster convergence over 1000 episodes.

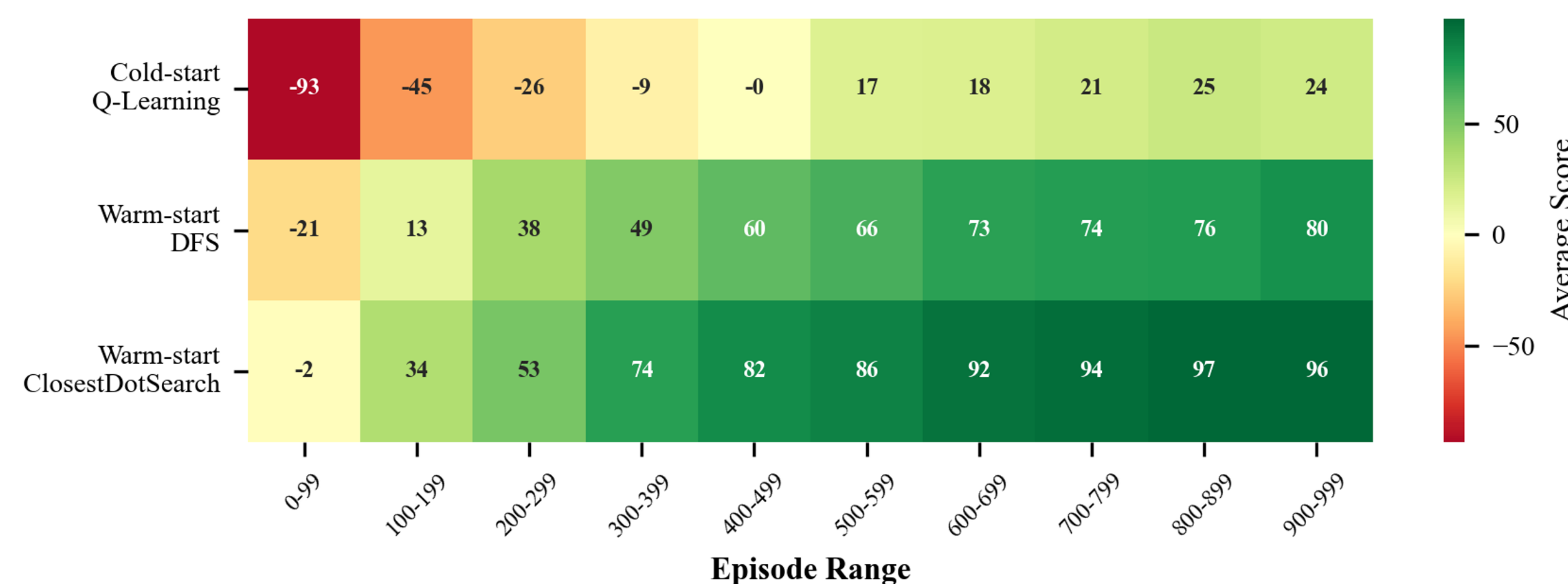


Fig. 2: Performance Heatmap Across Training

Average scores across episode ranges. Warm-start agents converge faster and perform more consistently, especially in early training.

Conclusion

Using suboptimal pathfinding to initialize Q-values provides a meaningful performance boost. The warm-start Q-learning agent converged faster, achieved higher scores, and won more frequently compared to the cold-start baseline. This demonstrates that demonstration-based initialization, even from imperfect or suboptimal trajectories, can effectively guide early exploration, reduce negative rewards, and accelerate the learning process in reinforcement learning tasks. Our results suggest that leveraging prior knowledge through demonstration not only improves learning efficiency but also results in more robust and higher-performing policies. This approach has practical implications for reinforcement learning in complex environments, where pure random exploration is inefficient and costly, and highlights the value of integrating search-based heuristics or demonstrations to bootstrap learning agents.

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